Thermocouple Measurement of Gas Stream using Lumped Capacitance Method

Saturday 10th October, 2026

Curvenote

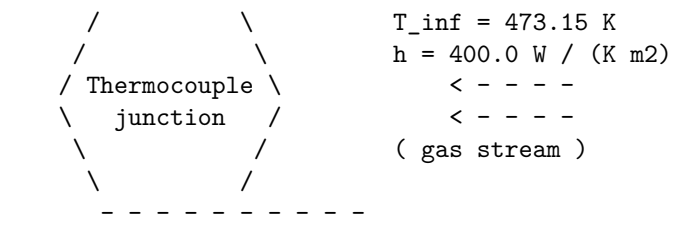
1 Problem Statement

2 Assumptions

- Temperature of junction is uniform at any instant
- Radiation exchange with the surroundings is negligible
- Losses by conduction through the leads are negligible
- Constant properties

3 Schematic

- T_i = 298.15 K
- k = 20 $\frac{\mathrm{W}}{\mathrm{K\,m}}$
- c = 400 $\frac{\mathrm{J}}{\mathrm{K\,kg}}$
- p = 8500 $\frac{\mathrm{kg}}{\mathrm{m^3}}$



4 Analysis

4.1 Thermocouple junction diameter

D = $\frac{6h\tau_t}{c\rho}$ = 0.70588235 mm

4.2 Thermocouple junction radius

r = $\frac{D}{2}$ = 0.35294118 mm

4.3 Characteristic length

L_c = $\frac{r}{3}$ = 0.11764706 mm

4.4 Biot number

Bi = $\frac{L_c h}{k}$ = 0.0023529412

4.5 time equation for lumped capacitance model

t = $-\tau_t \log\left(\frac{-T_{inf}+Tf}{T_i-T_{inf}}\right)$ = 5.164786 s

5 Comments

- lumped capacitance method is valid for Bi <0.1

6 References

- F. P. Incropera, Fundamentals of heat and mass transfer. 6th ed. John Wiley & Sons, 2007, Ch. 5, sec. 5.1-5.2, p.261-263.